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In situ Removal of Hydrogen Sulfide During Biogas Fermentation at Microaerobic Condition

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Abstract In this paper, rice straw was used as a raw material to produce biogas by anaerobic batch fermentation at 35 °C (mesophilic) or 55 °C (thermophilic). The hydrogen sulfide in biogas can be converted to S^0 or sulfate and removed in-situ under micro-oxygen environment. Trace oxygen was conducted to the anaerobic fermentation tank in amount of 0.5, 1.0, 2.0, 3.0, 4.0, 5.0, or 10.0 times stoichiometric equivalence, respectively, and the control experiment without oxygen addition was carried out. The results showed that the initial H_2S concentrations of biogas are about 3235 ± 185 mg/m^3 (mesophilic) or 3394 ± 126 mg/m^3 (thermophilic), respectively. The desulfurization efficiency is 72.3 % (mesophilic) or 65.6 % (thermophilic), respectively, with oxygen addition by stoichiometric relation. When the oxygen feeded in amount of 2~4 times, theoretical quantity demanded the removal efficiency of hydrogen sulfide could be over 92 %, and the oxygen residue in biogas could be maintained less than 0.5 %, which fit the requirement of biogas used as vehicle fuel or combined to the grid. Though further more oxygen addition could promote the removal efficiency of hydrogen sulfide (about 93.6 %), the oxygen residue in biogas would be higher than the application limit concentration (0.5 %). Whether mesophilic or thermophilic fermentation with the extra addition of oxygen, there were no obvious changes in the gas production and methane concentration. In conclusion, in-situ desulfurization can be achieved in the anaerobic methane fermentation system under micro-oxygen environment. In addition, air could be used as a substitute oxygen resource on the situation without strict demand for the methane content of biogas.

Keywords Biogas · Micro-oxygen · Desulfurization · Rice straw

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Introduction

Rice straw is very abundant in China and has long been an indispensable resource in Chinese rural life. The phenomenon of rice straw surplus has appeared with improvements in the rural energy structure and the use of alternative energy sources. Although the burning of rice straw in the field is forbidden, it does happen from time to time, which wastes valuable biomass resources and pollutes the atmosphere [1]. It makes sense to use rice straw as the raw material for biogas fermentation, thereby obtaining sustainable energy and avoiding environmental pollution [2].

Biogas is a kind of clean and renewable energy and could replace fossil energy, which would relieve the energy crisis and protect the environment. Biogas mainly comprises methane (60–70 %) and carbon dioxide (30–40 %) and also contains many other impurities such as hydrogen sulfide. Biogas can be used for power generation, as fuel for vehicles or incorporation into the natural gas network, but hydrogen sulfide concentration should be no more than 300, 15, and 20 mg/m^3, respectively [3, 4]. The concentration of hydrogen sulfide in initial biogas is 1–12 g/m^3 depending on the sulfide content of substrates [5], and it must be reduced to the above levels before use.

Wet and dry desulphurization is the primary traditional techniques for desulfurization of gasses. During wet desulfurization process, the hydrogen sulfide is removed through physical or chemical absorption while the gas countercurrent contact with specific sorbent [6]. The sorbent recycles after regeneration. Ammonia, sodium carbonate solution, organic amines, and other basic liquids are commonly used sorbents. High investment and cost, complex operation, and difficulty regeneration of used sorbent are main shortcomings [7, 8]. Dry desulfurization is realized by physical or chemical adsorption of hydrogen sulfide on some kinds of adsorbent such as iron oxide [9, 10]. The main difficulty is the regeneration of used adsorbent.

Biodesulfurization is a new technology, and specific microorganisms are needed to transfer hydrogen sulfide into elemental sulfur or sulfate [11, 12]. Light autotrophic microorganisms and microbial chemoautotroph are two key kinds of desulfurization microbial flora currently [13]. Soft sulfur bacteria are typical anaerobic autotrophic microorganisms which can use carbon dioxide to synthesize new cellular and transfer hydrogen sulfide in the presence of inorganic nutrients and illumination.

$$H_2S + CO_2 + O_2 + \text{nutrients} \overset{\text{microbial chemoautotrophs}}{\rightarrow} \text{biomass} + \text{S or } SO_4^{2-} + H_2O \quad (1)$$

The reaction of chemoautotroph microbial can be carried out under aerobic or anaerobic conditions under which oxygen work as an electron acceptor. While in the absence of oxygen, nitrites as an electron acceptor, hydrogen sulfide is transferred to elemental sulfur or sulfate. Theoretically, hydrogen sulfide can be directly oxidized into elemental sulfur or sulfate, the main reaction equation is as follows:

$$H_2S + \frac{1}{2}O_2 \rightarrow S + H_2O, \Delta G = -209.4kJ \Big/ mol \quad (2)$$

$$H_2S + 2O_2 \rightarrow SO_4^{2-} + 2H^+, \Delta G = -798.2kJ \Big/ mol \quad (3)$$

According to the above equations, Gibbs free energy (ΔG) is negative, which means the reactions can happen spontaneously. In other words, hydrogen sulfide in biogas can be easily

oxidized into elemental sulfur or sulfate in the presence of oxygen [9]. Obviously, excess oxygen is forbidden for anaerobic fermentation. However, in traditional biogas fermentation system, there is about 1 % facultative anaerobes which can protect strict anaerobic from micro-oxygen damage. So proper control of oxygen introduce may not only avoid the damage of anaerobe but also achieve the removal of hydrogen sulfide.

Biological desulfurization with limited oxygen supply has demonstrated a potential external process for the removal of H_2S from biogas [14]. However, the new biological system has to be performed in a additional equipment [15]. In fact, H_2S can react with oxygen spontaneously at room temperature and in-situ desulfurization could be obtained by introduction of trace oxygen into digestive system. Rencent report showed that H_2S in the biogas can be effectively removed [16]. The similar desulfurization effect can be achieved using air as the oxygen source, but the nitrogen in the air diluted the concentration of methane and reduced the calorific value of methane [17]. Hydrogen sulfide oxidation occurs mainly in the headspace of fermentation tank and the concentration of residual hydrogen sulfide in biogas can be controlled by regulation of oxygen supply [18]. It is necessary to enhance the understanding of a new technology of microaerobic desulfurization by further study. Moreover, the function of temperature, place and amount of oxygen added, and different oxygen sources need to be taken into consideration. In this paper, the experiments were carried out with rice straw as the raw material to investigate the feasibility of the direct addition of oxygen into the biogas fermentation chamber to achieve in situ desulfurization and the effects of technological parameters.

Material and Methods

Experimental Materials

The rice straw was collected from local farmland in Yuyao city of Zhejiang province, China. After washing and drying, the total solid (TS) content and volatile solid (VS) content of the straw were 91.0 and 85.2 %, respectively.

The inoculum was biogas slurry, which was obtained from the Zhengxing Animal Husbandry Limited Company, Hangzhou, China. The TS content and VS content were 0.57 and 0.47 %, respectively.

Experimental Design

The batch fermentation device comprised a glass biogas fermenting bottle (500 ml), a glass biogas collecting bottle (1000 ml), and a measuring cylinder.

The rice straw was cut into 2–3 cm size and crushed by the grinder. The rice straw and biogas slurry were put into the fermenting bottle successively, and the pH value was adjusted to 7. The fermenting bottle was sealed with rubber, and the residual air was replaced by argon to form anaerobic fermentation environment. The collecting bottle was first filled with water, and the pH value was regulated to 5 with sulfuric acid to reduce carbon dioxide dissolution later. The fermenting device was placed in a water bath controlled at 35 ± 1 °C (mesophilic fermentation) or 55 ± 1 °C (thermophilic fermentation).

The theoretical oxygen demand is defined as the oxygen needed to completely oxidize hydrogen sulfide in biogas into element sulfur, and it was calculated by the concentration

hydrogen sulfide in biogas and the biogas production according to the experiment in advance. Different times of the theoretical oxygen (0.5, 1.0, 2.0, 3.0, 4.0, 5.0, or 10.0) were introduced to the fermenting bottle, respectively. The experiments were repeated three times, and the control experiment without oxygen addition was also carried out.

Experimental Analytical Methods

Biogas was collected by drainage, and the daily biogas production was obtained with measuring cylinder. The TS and VS of rice straw or biogas slurry were measured using the gravimetric method. Biogas composition, such as oxygen and methane content, was determined by gas chromatography (GC, thermal conductivity detector (TCD), TDX-01column, carrier gas: argon, column temperature 90 °C, TCD temperature 120 °C, injection port temperature 150 °C). The hydrogen sulfide concentration in the biogas was measured using the "Natural gas - Determination of sulfur compound" standard (GB/T11060.2-2008) (Fig. 1).

Results and Discussion

Removal of Hydrogen Sulfide Under Different Oxygen Supplies

The removal efficiencies of hydrogen sulfide under different oxygen supply are shown in Fig. 2a, b and Table 1, representing mesophilic fermentation and thermophilic fermentation, respectively. As illustrated in Fig. 2, the hydrogen sulfide concentration in the biogas from the control experiment without oxygen addition showed small variances during the experimental period, and the average concentrations during mesophilic and thermophilic fermentation were 3235 ± 185 and 3394 ± 126 mg/m^3 , respectively, which could be used as a base to calculate desulfurization efficiency.

Figure 3 shows biogas oxygen residues content under various oxygen supplies (Fig. 3a, b represents mesophilic fermentation and thermophilic fermentation, respectively). When little amount of oxygen (0.5 time of theoretical demand) was supplied, the obvious reduction of hydrogen sulfide concentration for mesophilic or thermophilic fermentation was observed respectively, which explained that oxygen could react with hydrogen sulfide in anaerobic fermentation tank. The desulfurization efficiencies were relatively low (30.6 and 30.1 %,

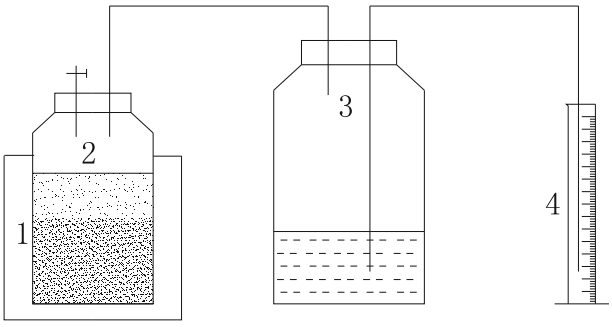
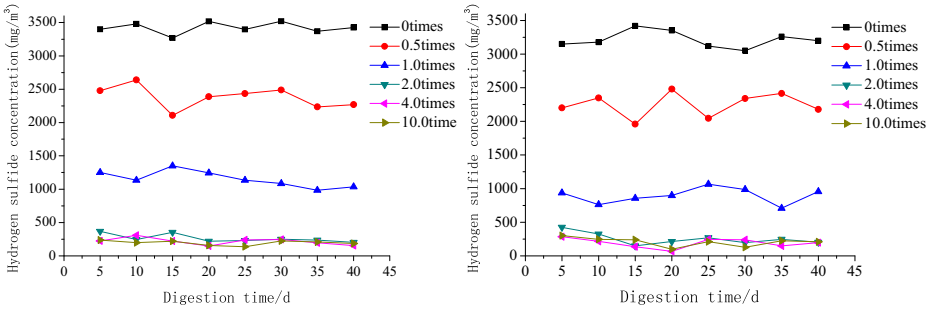


Fig. 1 Schematics of the batch fermentation device for microaerobic removal of hydrogen sulfide from biogas



(a) Fermentation temperature 35°C (b) Fermentation temperature 55°C

Fig. 2 Removal efficiency of hydrogen sulfide under various oxygen supplies. **a** Fermentation temperature 35 °C. **b** Fermentation temperature 55 °C

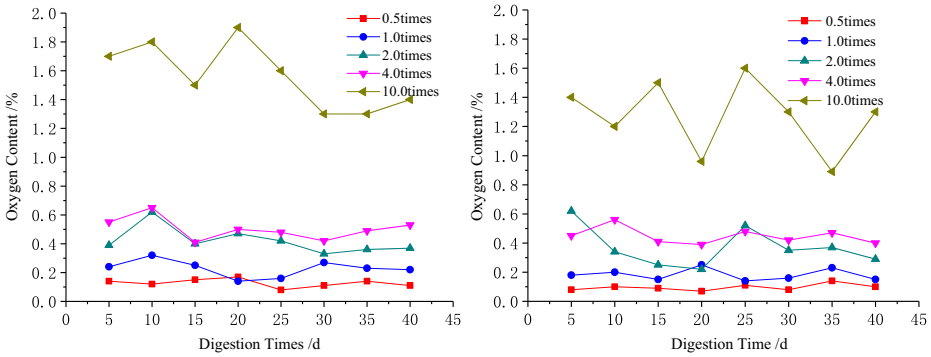
respectively), and the oxygen residue content was below 0.1 %. When the oxygen supply reached the theoretical oxygen demand, the average hydrogen sulfide concentration in the biogas significantly decreased to 800 ± 180 and 1167 ± 182 mg/m^3 for mesophilic and thermophilic fermentation, respectively. At the same time, the desulfurization efficiencies increased rapidly (72.3 and 65.6 %, respectively), and the oxygen residue content was still below 0.25 %. When oxygen was supplied at 2.0~4.0 times of theoretical demand, the average hydrogen sulfide concentration in the biogas during mesophilic and thermophilic fermentation were 166~314 and 248~285 mg/m^3, respectively, and the average removal efficiencies were 92.0~93.6 and 92.3~93.7 % for mesophilic and thermophilic fermentation, respectively. In this process, the hydrogen sulfide residue content was decreased with increased oxygen, which indicates appropriate excess oxygen will be needed. Although part hydrogen sulfide would be transferred into sulfate which consumes certain oxygen, it seems there are still superfluous oxygen entering bio-reaction system and wasted during the biogas fermentation process.

In Fig. 3, the concentration of hydrogen sulfide and oxygen stays below 500 mg/m^3 and 0.5 %, separately. When oxygen was added more than 4 times of theoretical oxygen demand, the average desulfurization efficiency during mesophilic and thermophilic fermentation was 93.8 and 94.5 %, respectively, but the oxygen residue content was above 0.5 %. And that superfluous oxygen will inhibit the activity of thiobacillus and hinder the oxidation process of hydrogen sulfide .

The above results show that the suitable oxygen supply was 2.0~4.0 times of theoretical demand under which the average desulfurization efficiency could be over 92 %, and the

Table 1 The relation of oxygen supply with H_2S removal/%

Oxygen supply (times)	H_2S average removal/%	
	35 °C	55 °C
0.5	31.4	30.5
1.0	72.6	66.3
2.0	92.3	92.3
4.0	94.2	93.7
10.0	93.8	94.5



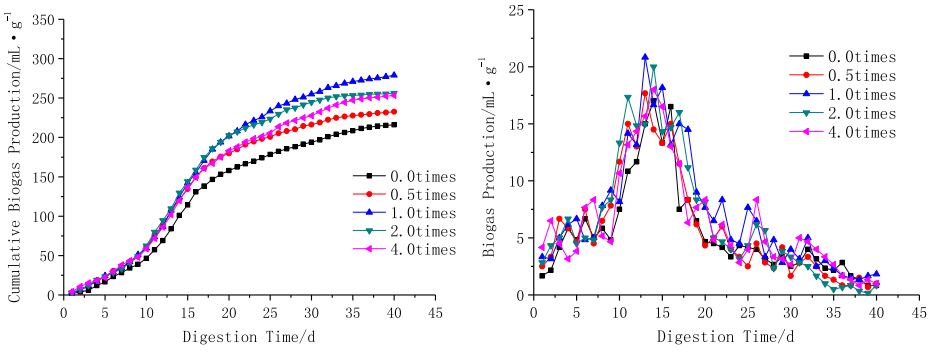
(a) Fermentation temperature 35°C (b) Fermentation temperature 55°C

Fig. 3 Oxygen residue under various oxygen supplies. **a** Fermentation temperature 35 °C. **b** Fermentation temperature 55 °C

oxygen residues content complied with the requirements of biogas use as car fuel or incorporated into the gas network (no more than 0.5 %).

Effect of Oxygen Addition on Biogas Production and Methane Content

Figure 4 shows the daily biogas production and cumulative biogas production under various oxygen supplies (mesophilic, Fig. 4a, b represents daily biogas production and cumulative biogas production, respectively). In the early stage (at the beginning of five days), biogas production level was relatively low and during this period hydrolysis, hydrogen production and acetic acid production had absolute predominance. Along with the fermentation process, methanogenic bacteria played a dominant role, mainly producing methane, while the production of biogas rapidly increases and the methane concentrations of biogas also gradually increases to a stable value of 60 %. As the fermentation continues, daily biogas production gradually decreased until no gas production and cumulative gas production also became gradually stable as the raw materials were expended and the generation of methane was attenuated. The addition of oxygen had some effect on daily biogas production and cumulative gas production, total biogas production all increased slightly compared with control



(a) cumulative biogas production (b) daily biogas production

Fig. 4 Cumulative biogas production and daily biogas production under various oxygen supplies. **a** Cumulative biogas production. **b** Daily biogas production

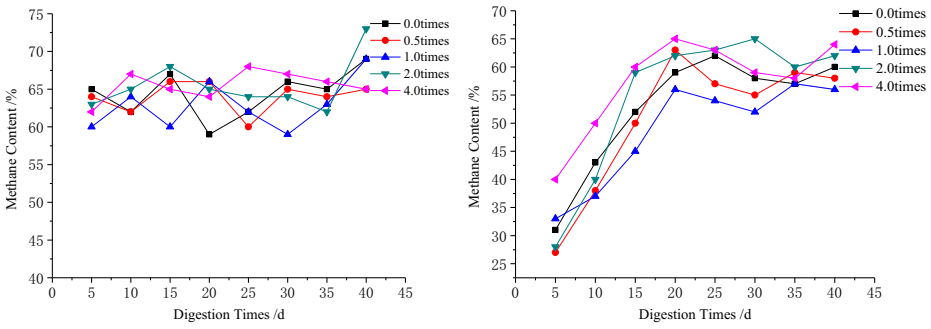
experiment (216.0 ml/g) without oxygen addition. These results indicate that the limited supply of oxygen did not have a significant effect on fermentation. The explanation of a small increase in gas production may be that limited oxygen supply causes facultative anaerobic bacteria in the fermentation system to use oxygen for hydrolysis which produced carbon dioxide or other gas.

Figure 5 shows the variations in methane concentration in fermentation flask under different oxygen supplies (Fig. 5a, b represents mesophilic and thermophilic fermentation, respectively). The methane content of biogas stabilized approximately at 60 % (35 °C) and 65 % (55 °C) though there were different oxygen supplies which explained the impact of oxygen supply to methane content not obvious. As shown in Fig. 5, methane concentration remained stable at 65 % in thermophilic fermentation, higher than mesophilic fermentation by about 5 %. The most probable explanation is that methanogens are more active under high temperature conditions in the middle stage when fermentation gave priority to the methane-producing phase. At high temperatures, the fermentation system can quickly enter the methanogenic stage. When oxygen was supplied with 0.5~1.0 time theoretical demand, the methane concentrations of biogas were slightly lower than control experiment, while they could be slightly higher when oxygen were supplied with 2.0~4.0 times theoretical demand. The explanation may be that the hydrogen sulfide concentration are greatly reduced if more oxygen being supplied, which reduced toxicity and increased methane production.

The Effects of Different Oxidants on Removal Efficiency of Hydrogen Sulfide

The hydrogen sulfide removal based the oxidation of hydrogen sulfide to elemental sulfur or sulfate. The air, containing 21 % oxygen, can be used as a source of oxygen to oxide hydrogen sulfide in the biogas. While air contains a lot of nitrogen which will affect the purity of biogas at least. The desulfurization experiments with rice straw as raw material and supplied with 1.0, 2.0, or 4.0 time of theoretical air demand, respectively, were carried out at high temperature (55 ± 1 °C) to explore the effect of different oxygen source on desulfurization efficiency, residual oxygen content, residual nitrogen content, and biogas production. The control experiment with oxygen as oxidant was done under the same condition.

As can be seen from Table 1, there were no obvious differences in biogas production and oxygen residues content when oxygen and air were chosen as the source of oxygen separately.



(a) Fermentation temperature 35°C (b) Fermentation temperature 55°C

Fig. 5 Methane concentration under various oxygen supply regimes. **a** Fermentation temperature 35 °C. **b** Fermentation temperature 55 °C

Because of the addition of air, nitrogen content in biogas increases evidently and methane content reduces slightly. When the air supply reached the theoretical oxygen demand, the desulfurization efficiency was 73.9 % and nitrogen content is 1.03 %. The desulfurization efficiency reached 93.9 %, and the nitrogen content could be controlled about 1.69 % when the air supply was 2.0 time of theoretical air demand. With the further increase of air addition, nitrogen content reach 2.64 %, which may influence the utilization of biogas, though the desulfurization efficiency continues to increase slightly. In conclusion, air is a kind of cheap and convenient choice as oxygen source in desulfurization of biogas for its utilization (such as combined heat and power (CHP)) (Table 2).

Conclusions

In this paper, hydrogen sulfide in the biogas was removed in situ by the micro-supply of oxygen during anaerobic batch fermentation with rice straw as raw material at 35 or 55 °C. The following conclusions can be drawn:

(1) There were no obvious changes in biogas production and methane concentration with the addition of limited oxygen in both mesophilic and thermophilic fermentation.
(2) When the oxygen supply was insufficient or just reached the theoretical oxygen demand, the desulfurization efficiencies were relatively low. When the oxygen supply was 2–4 times of the theoretical oxygen demand, the removal efficiency of hydrogen sulfide reaches 92 %, while the content of oxygen concentration is below 0.5 %. However, further increase of oxygen supply (such as 10 times the theoretical oxygen demand) would not promote the average removal efficiency obviously, while the content of oxygen residues in the biogas was much higher than the application limit (0.5 %).
(3) There were no obvious changes in biogas production and oxygen residues content when air was used as the source of oxygen but the nitrogen content is higher compared to the pure oxygen. The desulfurization efficiency reached 93.9 %, and the nitrogen content could be controlled about 1.69 % when the air supply was 2.0 time of theoretical air demand. At last a try to replace oxygen with air for the desulfurization was made, and the result shows that air could be used as oxygen source in desulfurization of biogas for its utilization (such as combined heat and power (CHP)).

Table 2 The experimental results of air or oxygen as oxygen sources

Oxygen supply	Oxygen source	H_2S content (mg/m^3)	H_2S removal efficiencies (%)	Oxygen content (%)	Nitrogen content (%)	Methane content (%)	biogas production (ml/g)	Methane production (ml/g)
1.0 times	air	886	74.1	0.22	1.03	62	295	183
	O_2	1153	66.3	0.23	0.09	63	292	184
2.0 times	air	208	93.9	0.45	1.69	63	317	200
	O_2	263	92.3	0.42	0.08	66	290	191
4.0 times	air	184	94.6	0.53	2.64	61	368	224
	O_2	217	93.7	0.50	0.08	66	336	222

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